

The logo for aiDAPTIV+ features the word "ai" in orange and "DAPTIV+" in black, with a plus sign. The background of the slide includes abstract blue and purple circuit-like patterns on the left and a futuristic robot head with a brain icon and a shield icon on the right.

aiDAPTIV+

架構與實作教學

歡迎參加本次 aiDAPTIV 技術教學課程。

本課程將深入探討 Phison 的 aiDAPTIV+ 解決方案，從其獨特的階層式記憶體架構到具體的實作細節，幫助您了解如何透過這項技術降低大型語言模型訓練的硬體門檻，並提升微調效率。

aiDAPTIVLink2 (Finetune)

1

階層式記憶體架構

aiDAPTIV+ 採用三級記憶體結構：
GPU VRAM (第一層) →
系統 DRAM (第二層) →
專用高耐久 SSD (第三層) 。

不活躍的模型資料可以動態「脫離」VRAM，
透過「換頁」機制移至 DRAM 或 SSD 上。

2

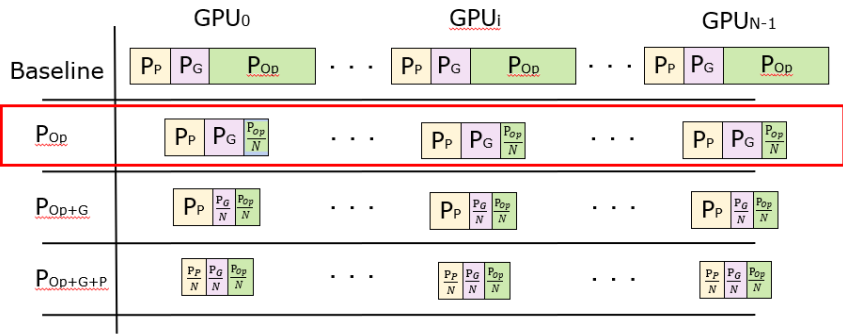
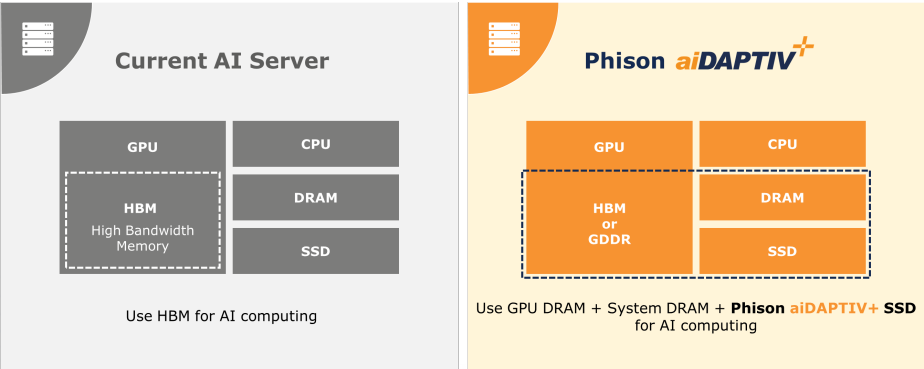
計算與記憶體解耦

透過專有的 Adaptive Link 中介軟體，
攔截 PyTorch 的記憶體請求，
改以 SSD 和系統DRAM來擴充 VRAM，
使 GPU 記憶體虛擬擴充數倍，且無需修改現有 AI
程式。

3

大幅降低硬體門檻

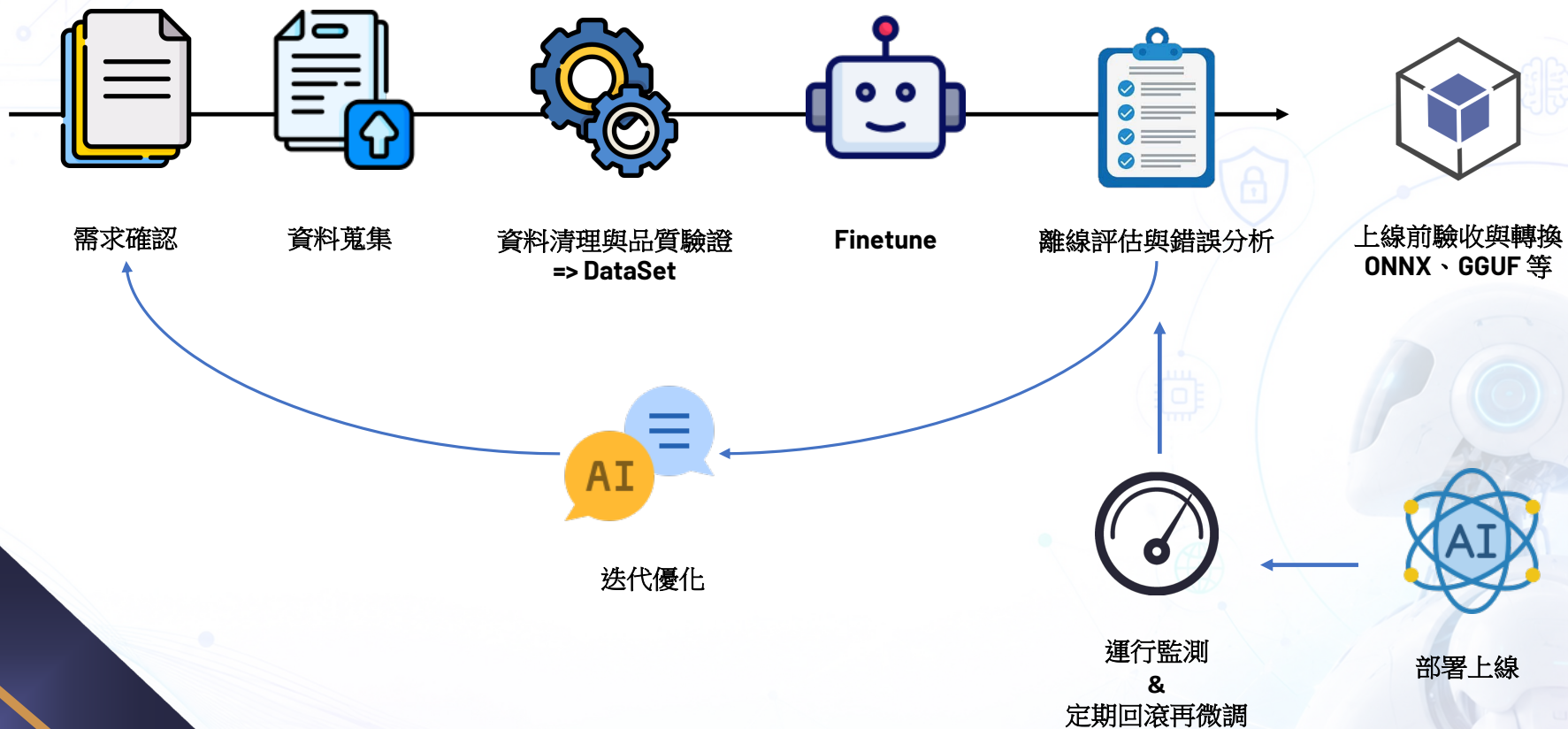
傳統需要 24 張 H100 GPU (約 1.4TB VRAM) 才能
運行 70B 模型，
而 aiDAPTIV+ 只需 4 張 RTX 6000 Ada GPU 加兩張
2TB aiDAPTIVCache SSD，
即可成功運行同樣規模的模型。



aiDAPTIV+ 提供了「直接取代式」的解決方案，讓中小企業、研究機構和教育單位無需投資昂貴的 GPU 陣列，即可在地端進行大型模型的微調。

單機系統可額外擴充至 320GB (PC 平台) 到 8TB (伺服器) 的可用記憶體，大幅提升 LLM 訓練的可行性與可負擔性。

aiDAPTIV+ 架構流程



Finetune — Dataset_config LLM (QA)

```
Open ▾ [🔍] QA_dataset_config.yaml ~/Desktop/aiDAPTIV_commands/dataset_config/text-generation

rm-static:
  data_path: "/user/Desktop/sample_instruction_data_1k_public.json"
  strategy: "QA"
  system_prompt: "This is a chat between a user and an artificial intelligence assistant. The assistant gives helpful,
detailed, and polite answers to the user's questions based on the context. The assistant should also indicate when the answer
cannot be found in the context."
  user_prompt: "{question}"
  question_key: "instruct"
  answer_key: "output"
  exp_type: train ## train or inference or eval
  label_key: "output" ## same as answer key
```

Finetune — Dataset_config LLM (Pretrain)



The image shows a code editor window with the title "Pretrain_dataset_config.yaml" and a file path "~/Desktop/aiDAPTIV_commands/dataset_config/text-generation". The editor contains a YAML configuration for pretraining an LLM. The configuration includes fields for data path, strategy, system and user prompts, text type, and experiment type. The strategy is set to "Pretrain" and the experiment type is set to "train".

```
Pretrain_dataset:
  data_path: "upstage/Pretraining_Dataset"
  strategy: "Pretrain"
  system_prompt: ""
  user_prompt: ""
  text: "text"
  exp_type: train ## train or inference or eval
```


Finetune — Dataset_config LLM (RAG)

```
Open ▾  [🔍]  RAG_dataset_config.yaml  ~/Desktop/aiDAPTIV_commands/dataset_config/text-generation  ⚙️  ☰  -  🖼️  ✕
```

```
RAG_dataset:
  data_path: "neural-bridge/rag-dataset-12000"
  strategy: "RAG"
  system_prompt: "This is a chat between a user and an artificial intelligence assistant. The assistant gives helpful,
detailed, and polite answers to the user's questions based on the context. The assistant should also indicate when the answer
cannot be found in the context."
  user_prompt: "Question:{question}, Context:{rag}"
  question_key: "question"
  answer_key: "answer"
  rag_key:
    - "context"
  exp_type: train ## train or inference or eval
  label_key: "answer"  ## same as answer key
```

Finetune — Env_config LLM

```
Open  ▾  [1]  llm_env_config.yaml
~/Desktop/aiDAPTIV_commands/env_config

# Save_path, nvme_path, log_name settings
path_settings:
  lora:
    lora_weight: ""          # whether to load lora_weight (only activated when lora: true)
    lora_optimizer: ""       # whether to load lora_optimizer (only activated when lora: true)
    lora_output_dir: ""      # whether to save lora adapter weight (only activated when lora: true)
  model_name_or_path: "/app/Meta-Llama-3.1-8B-Instruct"
  multi_node_env_path: null
  optimizer_path: "" # whether to load optimizer
  train_data_path:
    # - ./dataset_config/automatic-speech-recognition/dataset_config.yaml
    # - ./dataset_config/image-text-to-text/VQA_dataset_config.yaml
    # - ./dataset_config/text-generation/Pretrain_dataset_config.yaml
    # - ./dataset_config/text-generation/RAG_dataset_config.yaml
    - /user/Desktop/aiDAPTIV_commands/dataset_config/text-generation/QA_dataset_config.yaml
  val_data_path: null
    # - ./dataset_config/automatic-speech-recognition/dataset_config.yaml
    # - ./dataset_config/image-text-to-text/VQA_dataset_config.yaml
    # - ./dataset_config/text-generation/Pretrain_dataset_config.yaml
    # - ./dataset_config/text-generation/RAG_dataset_config.yaml
    # - ./dataset_config/text-generation/QA_dataset_config.yaml
  nvme_path: "/mnt/nvme0"
  output_dir: "/user/Desktop/output"
  log_name: "llm_test"
```

Finetune — Exp_config LLM

Open ▾

llm_exp_config.yaml

~/Desktop/aiDAPTIV_commands/exp_config

```
# Environmental settings
process_settings:
  num_gpus: 1
  specify_gpus: '0'
  master_port: 8299
  multi_node_settings:
    enable: False
    master_addr: "127.0.0.1"

# Running train, eval, inference related settings
run_settings:
  task_type: "text-generation"
  task_mode: "train"
  per_device_train_batch_size: 1
  per_update_total_batch_size: 4
  num_train_epochs: 1
  max_iter: -1
  max_seq_len: 2048
  triton: True
  weight_file_format: null
  from_config: false
  precision_mode: 1
  enable_save_optimizer_state: false
```


Finetune — Exp_config LLM

model_saver:

```
max_num_of_saved_model_on_epoch_end: -1 # If the value is -1, it is equal to num_train_epochs.  
enable_save_model_on_iteration: false  
max_num_of_saved_model_on_iteration: 2  
num_of_iteration_to_save_model: 2
```

lr_scheduler:

```
mode: 1  
learning_rate: 0.000007
```

optimizer:

```
beta1: 0.9  
beta2: 0.95  
eps: 0.00000001  
weight_decay: 0.01
```

Finetune — Exp_config LLM

Open ▾

llm_exp_config.yaml
~/Desktop/aiDAPTIV_commands/exp_config

```
early_stop:
  enable: false
  min_delta: 0.01
  patience: 2
  verbose: false

lora:
  enable_lora: false
  lora_rank: 8
  lora_alpha: 16
  lora_task_type: "CAUSAL_LM"
  lora_target_modules: null
```

Check docker image list

```
user@user-E500-G9:~$ sudo docker image list
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
aidaptiv:vnXUN_2_04_A1	a19b7faf2e1c	9.84GB	0B	U
aidaptiv:vnXUN_3_03_AA	a041e3c866d7	12.1GB	0B	U

Demo LLM

1. 啟動 Docker Container

執行以下指令建立並啟動容器環境：

```
sudo docker run --gpus all -it --name phison_MW204_test  
--ipc=host --privileged=true --ulimit memlock=-1 --ulimit  
stack=67108864 -v /usr/local/models:/app -v  
/mnt/nvme0:/mnt/nvme0 -v /dev/mapper:/dev/mapper -v  
/var/lock:/var/lock -v /home/user:/user  
aidaptiv:vnXUN_2_04_A1
```

- -it：可進入 container 操作 bash
- --gpus all：將所有 GPU 提供給 Container 使用
- --ipc=host：共享 Host 的 IPC
- --privileged=true：給 Container 高權限模式
- --ulimit memlock=-1：取消 Memory Lock 限制
- --ulimit stack=67108864：增加 Stack Size，提供較大的執行緒堆疊空間
- -v：Host 與 Container 共用資料

Demo LLM

2. 確認容器啟動

成功進入容器後，會看到以下提示：

```
user@user-Pro-ET500I-W8:~/Desktop$ sudo docker run --gpus all -it --name  
k=-1 --ulimit stack=67108864 -v /usr/local/models:/app -v /mnt/nvme0:/mnt  
ome/user:/user aidaptiv:VNXUN_2_04_A1  
root@9bfb6ddc4404:/#
```


Demo LLM

3. 確認vNXUN204.A1

```
root@9bfb6ddc4404:/user/Desktop# phisonai2 --version
phisonai2: Phison aiDAPTIV aiDAPTIVLink
COPYRIGHT (C) 2024 Phison Electronics Corp.
version: vNXUN204.A1
```

Demo LLM

4. 執行Finetune

```
root@9bfb6ddc4404:/user/Desktop# phisonai2 --env_config /user/Desktop/aiDAPTIV_commands/env_config/llm_env_config.yaml --exp_config /user/Desktop/aiDAPTIV_commands/exp_config/llm_exp_config.yaml
[rank0]:[W525 07:48:12.904352125 ProcessGroupNCCL.cpp:4715] [PG ID 0 PG G
UID 0 Rank 0] using GPU 0 as device used by this process is currently un
known. This can potentially cause a hang if this rank to GPU mapping is i
ncorrect. You can pecify device_id in init_process_group() to force use o
f a particular device.
You are attempting to use Flash Attention 2.0 with a model not initialize
d on GPU. Make sure to move the model to GPU after initializing it on CPU
with `model.to('cuda')`.
Loading checkpoint shards: 100%|██████████| 4/4 [00:00<00:00, 19.21it/s]
```

Demo LLM

5. 查看Logs

```
user@user-Pro-ET500I-W8:~/Desktop$ tail llm_test -f
dist_world_size=1
Setting CUDA_VISIBLE_DEVICES=0
#### [INITIALIZE Tokenizer] ===== ####
#### [INITIALIZE Dataloader] ===== ####
[INFO] get dataset name: /user/Desktop/sample_instruction_data_1k_public.json training_strategy: Q
A
#### [INITIALIZE Model] ===== ####
[INFO] [create_hf_model] init from pretrained
[INFO] You are loading [LlamaForCausalLM] with FlashAttention2
[INFO] You are setting [LlamaForCausalLM] with Gradient Checkpointing
[INFO] create current swap data path: /mnt/nvme0/phison_99
[INFO] You are accelerating [LlamaForCausalLM] with FusedEmbedding kernel
[INFO] You are accelerating [LlamaForCausalLM] with FusedGateMLP kernel
[INFO] You are accelerating [LlamaForCausalLM] with FusedQKVOLinear kernel
[INFO] You are accelerating [LlamaForCausalLM] with FusedRMSNorm kernel
[INFO] You are accelerating [LlamaForCausalLM] with FusedRoPE kernel
[INFO] You are accelerating [LlamaForCausalLM] with FusedLinearXentropy kernel
[2026-05-25 07:49:20.971786] [INFO] Enable Early-stopping: False
[INFO] The current training schedule consists of 1 epochs with 1000 iterations each.
[INFO] The accumulated time is set to 4 so the training will be updated 250 times during each epoch.
[INFO] If the trained model does not show significant improvement, you may consider increasing the number of update cycles.
[INFO] If the trained model shows signs of overfitting, you should decrease the number of update cycles.
[2026-05-25 07:49:20.971860] [INFO] Beginning of Epoch 1/1, Total Micro Batches 1000
[2026-05-25 07:49:20.996540] [PHISON START] Epoch: 0, Iteration: 0
[2026-05-25 07:49:20.997441] [Forward][start]
[2026-05-25 07:49:26.884343] [Forward][time spent]:5.886889457702637
[2026-05-25 07:49:27.114136] [Loss]:2.8514997959136963
[2026-05-25 07:49:27.114153] [Backward][start]
[2026-05-25 07:49:34.632935] [Backward][time spent]:7.518741130828857
[2026-05-25 07:49:34.632993] [Update][Start]
[2026-05-25 07:49:34.913905] [Update][time spent]:0.2809011936187744
[2026-05-25 07:49:34.913971] [PHISON END] Iteration: 0
```

Demo LLM

5. 查看Logs

```
[2026-05-25 07:49:34.915726] [PHISON START] Epoch: 0, Iteration: 1
[2026-05-25 07:49:34.915818] [Forward][start]
[2026-05-25 07:49:37.435608] [Forward][time spent]:2.51977801322937
[2026-05-25 07:49:37.672774] [Loss]:2.059720754623413
[2026-05-25 07:49:37.672786] [Backward][start]
[2026-05-25 07:49:48.078951] [Backward][time spent]:10.406155347824097
[2026-05-25 07:49:48.079017] [Update][Start]
[2026-05-25 07:49:48.607173] [Update][time spent]:0.5281453132629395
[2026-05-25 07:49:48.607241] [PHISON END] Iteration: 1

[2026-05-25 07:49:48.609082] [PHISON START] Epoch: 0, Iteration: 2
[2026-05-25 07:49:48.609184] [Forward][start]
[2026-05-25 07:49:51.126857] [Forward][time spent]:2.5176608562469482
[2026-05-25 07:49:51.363920] [Loss]:3.0969350337982178
[2026-05-25 07:49:51.363932] [Backward][start]
[2026-05-25 07:50:01.753068] [Backward][time spent]:10.38912582397461
[2026-05-25 07:50:01.753136] [Update][Start]
[2026-05-25 07:50:02.270922] [Update][time spent]:0.5177762508392334
[2026-05-25 07:50:02.270989] [PHISON END] Iteration: 2

[2026-05-25 07:50:02.272686] [PHISON START] Epoch: 0, Iteration: 3
[2026-05-25 07:50:02.272793] [Forward][start]
[2026-05-25 07:50:04.789683] [Forward][time spent]:2.516878366470337
[2026-05-25 07:50:05.026719] [Loss]:2.533737897872925
[2026-05-25 07:50:05.026731] [Backward][start]
[2026-05-25 07:50:15.498895] [Backward][time spent]:10.472153902053833
[2026-05-25 07:50:15.498957] [Update][Start]
using avx2
max_compute_thread = 18
[2026-05-25 07:50:58.376221] [Update][time spent]:42.877254009246826
[2026-05-25 07:50:58.376315] [PHISON END] Iteration: 3

[2026-05-25 07:50:58.376331] [INFO] Training efficiency: 84.10291583774017 (tokens/s)

[2026-05-25 07:50:58.382439] [PHISON START] Epoch: 0, Iteration: 4
[2026-05-25 07:50:58.385697] [Forward][start]
```

Finetune — Dataset_config VLM (QA)

```
Open ▾  [+]
```

VQA_dataset_config.yaml
~/Desktop/aiDAPTIV_commands/dataset_config/image-text-to-text

```
QA_dataset:
  data_path: "/user/Desktop/MathVista"
  image_folder : "/user/Desktop/MathVista"
  strategy: "QA"
  system_prompt: "This is a chat between a user and an artificial intelligence assistant. The assistant gives helpful,
detailed, and polite answers to the user's questions based on the context. The assistant should also indicate when the answer
cannot be found in the context."
  user_prompt: "is the description fits the {question}"
  question_key: "question"
  answer_key: "answer"
  image_key : "image"
  exp_type: train ## train or inference
  label_key: "answer"
```


Finetune — Env_config VLM

Open ▾

vlm_env_config.yaml

~/Desktop/aiDAPTIV_commands/env_config

```
# Save_path, nvme_path, log_name settings
path_settings:
  lora:
    lora_weight: ""      # whether to load lora_weight (only activated when lora: true)
    lora_optimizer: ""   # whether to load lora_optimizer (only activated when lora: true)
    lora_output_dir: ""  # whether to save lora adapter weight (only activated when lora: true)
  model_name_or_path: "/app/Llama-3.2-11B-Vision-Instruct"
  multi_node_env_path: null
  optimizer_path: "" # whether to load optimizer
  train_data_path:
    # - ./dataset_config/automatic-speech-recognition/dataset_config.yaml
    # - /user/Desktop/aiDAPTIV_commands/dataset_config/image-text-to-text/VQA_dataset_config.yaml
    # - ./dataset_config/text-generation/Pretrain_dataset_config.yaml
    # - ./dataset_config/text-generation/RAG_dataset_config.yaml
    # - ./dataset_config/text-generation/QA_dataset_config.yaml
  val_data_path: null
    # - ./dataset_config/automatic-speech-recognition/dataset_config.yaml
    # - ./dataset_config/image-text-to-text/VQA_dataset_config.yaml
    # - ./dataset_config/text-generation/Pretrain_dataset_config.yaml
    # - ./dataset_config/text-generation/RAG_dataset_config.yaml
    # - ./dataset_config/text-generation/QA_dataset_config.yaml
  nvme_path: "/mnt/nvme0"
  output_dir: "/user/Desktop/output"
  log_name: "vlm_test"
```

Finetune — Exp_config VLM

Open ▾ 

vlm_exp_config.yaml

~/Desktop/aiDAPTIV_commands/exp_config

vlm_exp_config.yaml

```
# Environmental settings
process_settings:
  num_gpus: 1
  specify_gpus: null
  master_port: 8299
  multi_node_settings:
    enable: False
    master_addr: "127.0.0.1"

# Running train, eval, inference related settings
run_settings:
  task_type: "image-text-to-text"
  task_mode: "train"
  per_device_train_batch_size: 1
  per_update_total_batch_size: 4
  num_train_epochs: 1
  max_iter: -1
  max_seq_len: 2048
  triton: True
  weight_file_format: null
  from_config: false
  precision_mode: 1
  enable_save_optimizer_state: false
```



Finetune — Exp_config VLM

Open ▾

vlm_exp_config.yaml

~/Desktop/aiDAPTIV_commands/exp_config

```
model_saver:
  max_num_of_saved_model_on_epoch_end: -1 # If the value is -1, it is equal to num_train_epochs.
  enable_save_model_on_iteration: false
  max_num_of_saved_model_on_iteration: 2
  num_of_iteration_to_save_model: 2

lr_scheduler:
  mode: 1
  learning_rate: 0.000007

optimizer:
  beta1: 0.9
  beta2: 0.95
  eps: 0.00000001
  weight_decay: 0.01
```

Finetune — Exp_config VLM

```
early_stop:  
  enable: false  
  min_delta: 0.01  
  patience: 2  
  verbose: false  
  
lora:  
  enable_lora: false  
  lora_rank: 8  
  lora_alpha: 16  
  lora_task_type: "CAUSAL_LM"  
  lora_target_modules: null
```

Demo VLM

4. 執行Finetune

```
root@9bfb6ddc4404:/user/Desktop# phisonai2 --env_config /user/Desktop/aiDAPTIV_commands/env_config/vlm_env_config.yaml --exp_config /user/Desktop/aiDAPTIV_commands/exp_config/vlm_exp_config.yaml
Generating testmini split: 100%|██████████| 1000/1000 [00:00<00:00, 2432.00 examples/s]
Generating test split: 100%|██████████| 5141/5141 [00:01<00:00, 3101.67 examples/s]
[rank0]:[W525 09:13:57.388864847 ProcessGroupNCCL.cpp:4715] [PG ID 0 PG GUID 0 Rank 0] using GPU 0
as device used by this process is currently unknown. This can potentially cause a hang if this ran
k to GPU mapping is incorrect. You can pecify device_id in init_process_group() to force use of a p
articular device.
Loading checkpoint shards: 100%|██████████| 5/5 [00:00<00:00, 7.60it/s]
```

4. 執行Finetune

```
root@9bf6ddc4404:/user/Desktop# phisonai2 --env_config /user/Desktop/aiDAPTIV_commands/env_config/vlm_env_config.yaml --exp_config /user/Desktop/aiDAPTIV_commands/exp_config/vlm_exp_config.yaml
Generating testmini split: 100%|██████████| 1000/1000 [00:00<00:00, 2432.00 examples/s]
Generating test split: 100%|██████████| 5141/5141 [00:01<00:00, 3101.67 examples/s]
[rank0]:[W525 09:13:57.388864847 ProcessGroupNCCL.cpp:4715] [PG ID 0 PG GUID 0 Rank 0] using GPU 0
as device used by this process is currently unknown. This can potentially cause a hang if this ran
k to GPU mapping is incorrect. You can pecify device_id in init_process_group() to force use of a p
articular device.
Loading checkpoint shards: 100%|██████████| 5/5 [00:00<00:00, 7.60it/s]
```


Demo LLM

5. 查看Logs

```
user@user-Pro-ET500I-W8:~/Desktop$ tail vlm_test -f
dist_world_size=1
Setting CUDA_VISIBLE_DEVICES=0
#### ===== [INITIALIZE Tokenizer] ===== ####
#### ===== [INITIALIZE Dataloader] ===== ####
#### ===== [INITIALIZE Model] ===== ####
[INFO] [create_hf_model] init from pretrained
[INFO] [MllamaForConditionalGeneration] does not support FlashAttention2
[INFO] You are setting [MllamaForConditionalGeneration] with Gradient Checkpointing
mllama don't resize token_embeddings to avoid modifying config.text_config.vocab_size
[INFO] create current swap data path: /mnt/nvme0/phison_368
[INFO] You are accelerating [MllamaForConditionalGeneration] VisionEncoder with FusedEmbedding kernel
[INFO] You are accelerating [MllamaForConditionalGeneration] VisionEncoder with FusedMLP kernel
[INFO] You are accelerating [MllamaForConditionalGeneration] VisionEncoder with FusedQKVOLinear kernel
[INFO] You are accelerating [MllamaForConditionalGeneration] VisionEncoder with FusedLayerNorm kernel
[INFO] You are accelerating [MllamaForConditionalGeneration] VisionEncoder with FlashAttention2
[INFO] You are accelerating [MllamaForConditionalGeneration] TextDecoder with FusedEmbedding kernel
[INFO] You are accelerating [MllamaForConditionalGeneration] TextDecoder with FusedGateMLP kernel
[INFO] You are accelerating [MllamaForConditionalGeneration] TextDecoder with FusedQKVOLinear kernel
[INFO] You are accelerating [MllamaForConditionalGeneration] TextDecoder with FusedRMSNorm kernel
[INFO] You are accelerating [MllamaForConditionalGeneration] TextDecoder with FusedRoPE kernel
[INFO] You are accelerating [MllamaForConditionalGeneration] TextDecoder with FusedLinearXentropy kernel
[INFO] You are accelerating [MllamaForConditionalGeneration] TextDecoder with FlashAttention2
[2026-05-25 09:15:30.080234] [INFO] Enable Early-stopping: False
[INFO] The current training schedule consists of 1 epochs with 1000 iterations each.
[INFO] The accumulated time is set to 4 so the training will be updated 250 times during each epoch.
[INFO] If the trained model does not show significant improvement, you may consider increasing the number of update cycles.
[INFO] If the trained model shows signs of overfitting, you should decrease the number of update cycles.
Beginning of Epoch 1/1, Total Micro Batches 1000
[2026-05-25 09:15:30.200903] [PHISON START] Epoch: 0, Iteration: 0
[2026-05-25 09:15:30.203315] [Forward][start]
[2026-05-25 09:15:39.984466] [Forward][time spent]:9.781121492385864
[2026-05-25 09:15:40.237445] [Loss]:13.156944274902344
[2026-05-25 09:15:40.237459] [Backward][start]
[2026-05-25 09:15:53.289252] [Backward][time spent]:13.051785230636597
[2026-05-25 09:15:53.289314] [Update][Start]
[2026-05-25 09:15:53.421476] [Update][time spent]:0.13215398788452148
```

Demo LLM

5. 查看Logs

```
[2026-05-25 09:15:53.463984] [PHISON START] Epoch: 0, Iteration: 1
[2026-05-25 09:15:53.464760] [Forward][start]
[2026-05-25 09:15:58.189621] [Forward][time spent]:4.724838972091675
[2026-05-25 09:15:58.445017] [Loss]:12.912070274353027
[2026-05-25 09:15:58.445029] [Backward][start]
[2026-05-25 09:16:14.623016] [Backward][time spent]:16.177977800369263
[2026-05-25 09:16:14.623080] [Update][Start]
[2026-05-25 09:16:14.755378] [Update][time spent]:0.13228106498718262
[2026-05-25 09:16:14.755443] [PHISON END] Iteration: 1

[2026-05-25 09:16:14.799349] [PHISON START] Epoch: 0, Iteration: 2
[2026-05-25 09:16:14.800127] [Forward][start]
[2026-05-25 09:16:19.520171] [Forward][time spent]:4.720027208328247
[2026-05-25 09:16:19.776926] [Loss]:13.038264274597168
[2026-05-25 09:16:19.776941] [Backward][start]
[2026-05-25 09:16:35.957746] [Backward][time spent]:16.180795431137085
[2026-05-25 09:16:35.957805] [Update][Start]
[2026-05-25 09:16:36.088206] [Update][time spent]:0.13039350509643555
[2026-05-25 09:16:36.088270] [PHISON END] Iteration: 2

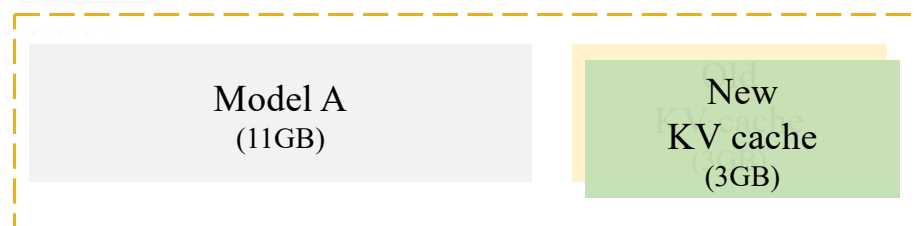
[2026-05-25 09:16:36.144715] [PHISON START] Epoch: 0, Iteration: 3
[2026-05-25 09:16:36.145484] [Forward][start]
[2026-05-25 09:16:40.869894] [Forward][time spent]:4.7243876457214355
[2026-05-25 09:16:41.126688] [Loss]:13.015402793884277
[2026-05-25 09:16:41.126699] [Backward][start]
[2026-05-25 09:16:57.397445] [Backward][time spent]:16.27073907852173
[2026-05-25 09:16:57.397507] [Update][Start]
using avx2
max_compute_thread = 18
[2026-05-25 09:17:51.118156] [Update][time spent]:53.72063946723938
[2026-05-25 09:17:51.118235] [PHISON END] Iteration: 3

[Training efficiency] Image + Text : 239.71 (tokens/s)
[2026-05-25 09:17:51.144823] [PHISON START] Epoch: 0, Iteration: 4
[2026-05-25 09:17:51.334856] [Forward][start]
[2026-05-25 09:17:56.050707] [Forward][time spent]:4.7158122062683105
[2026-05-25 09:17:56.307527] [Loss]:11.47286605834961
```

aiDAPTIVLink3 (Inference)

Reuse KV cache

Without aiDAPTIV+



Unified Memory



Re-compute

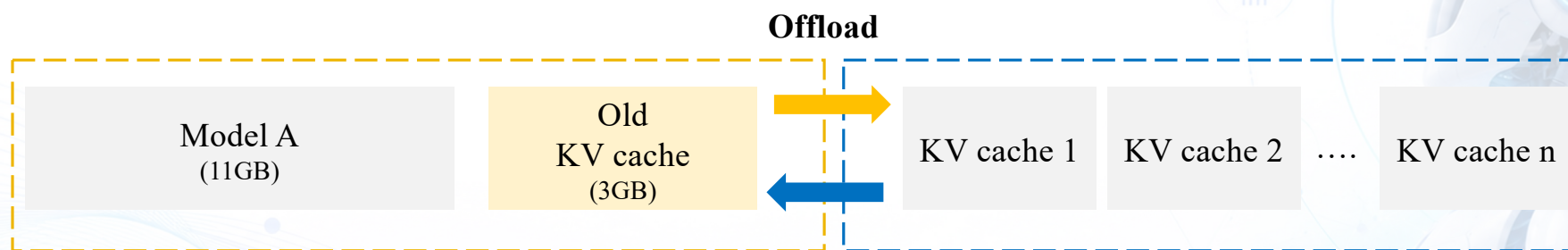
Example

GenAI Application in GPU

VRAM Memory Size : 16GB

Model A Size : 11GB (Qwen 2.5 14B INT4)

Model A KV cache Size : 3GB



Unified Memory

Load

External Memory (SSD)

aiDAPTIVLink3 (Inference)

KV Cache offload 機制最核心的價值所在——它本來就不是為了「100% 完全相同」設計的，而是為了「部分重複」設計的。系統不需要你的 Request 跟之前「完全一樣」，採用（**Prefix Matching**）機制，也就是「開頭長得一樣就可以」。

第一步：

假設您第一次送出的 Request 是這樣（並已成功 Lock 在 Doc1 資料夾）：

[System: 你是硬體架構師] **[SPEC文件全文：USB PD 3.0 規範.....(5000字)]** **[使用者的問題：請問這份規範的最高電壓]**
系統會把這整串文字切成 **Token (標記)**，並儲存成一個「前綴樹」：

第二步：

過 5 分鐘後，另一位同事想問同份 Spec 的不同問題：

[System: 你是硬體架構師] **[SPEC文件全文：USB PD 3.0 規範.....(5000字)]** **[使用者的問題：這份規範支援多少瓦的功]**
比對邏輯：
系統從第一個 Token 「你」開始比對，一路比對到「請問」全都一模一樣。直到第一個不同的開始分支。

跳過前面超大token的運算，只針對「請問...之後的內容」這後半段進行全新的計算。

➡ 提速效果顯著（節省約 50%~70% 運算）。

新算出來的「電壓問題」、「功率問題」快取，被系統丟進動態（**Dynamic**）區域。
日後若硬碟滿了，這組「Spec + 電壓」和「Spec + 功率」的快取可能會被刪除。

Demo aiDAPTIVLink3.0

1. 啟動 Docker Container

執行以下指令建立並啟動容器環境：

```
sudo docker run -d --gpus all --name phison_MW303_test --  
ipc=host --pid=host --privileged=true --ulimit memlock=-1 --  
ulimit stack=67108864 -p 8000:8000 -v  
/usr/local/models:/app -v /mnt/nvme0:/mnt/nvme0 -v  
/dev/mapper:/dev/mapper aidaptiv:vNXUN_3_03_AA
```

- -d：背景（Detach）模式執行
- -it：可進入 container 操作 bash
- --gpus all：將所有 GPU 提供給 Container 使用
- --ipc=host：解決深度學習 DataLoader 共享記憶體不足（/dev/shm 太小）的問題。
- --pid=host：讓容器內行程管理工具能看見主機所有行程，便於除錯、發送信號及 GPU 佔用追蹤。
- --privileged=true：給 Container 高權限模式
- --ulimit memlock=-1：深度學習框架（如 PyTorch/TensorFlow）常用 cudaMalloc 或 pinned memory（頁面鎖定記憶體）來加速 GPU 資料傳輸，設定為 -1 允許容器鎖住所有實體記憶體
- --ulimit stack=67108864：增加 Stack Size，提供較大的執行緒堆疊空間
- -p 8000:8001：Port Mapping，
- Host 的 8000 Port 對應到 Container 的 8000 Port
- -v /dev/mapper:/dev/mapper：Linux 的 Device Mapper 目錄，掛載這個才能讓容器內直接存取底層的區塊設備或掛載的儲存池。

Demo LLM

2. 確認容器啟動

成功進入容器後，會看到以下提示：(cd /home/root/aiDAPTIV2/script/)

```
user@user-Pro-ET500I-W8:~$ sudo docker run -d --gpus all -it --name phison_MW303_test --ipc=host --pid=host --privileged=true --ulimit memlock=-1 --ulimit stack=67108864 -p 8000:8000 -v /usr/local/models:/app -v /mnt/nvme0:/mnt/nvme0 -v /dev/mapper:/dev/mapper aidaptiv:vNXUN_3_03_AA 2d272c6f3064895df3a440cfc91d9191504a93b4f9feaeb93aae0120b85d88d7
user@user-Pro-ET500I-W8:~$ sudo docker exec -it 2d2 bash
root@2d272c6f3064:/# cd /home/root/aiDAPTIV2/script/
root@2d272c6f3064:/home/root/aiDAPTIV2/script#
```

Demo aiDAPTIVLink3.0

3. 啟用 Model

執行以下指令啟用模型：

```
VLLM_USE_V1=1
VLLM_WORKER_MULTIPROC_METHOD=spawn python3
vllm_api_server.py --model /app/gemma-3-12b-it/ --served-
model-name gemma-3-12b-it --nvme_path
/mnt/nvme0/inference --gpu-memory-utilization 0.8 --max-
model-len 60000 --tensor-parallel-size 2 --enable-chunked-
prefill --enable-prefix-caching --disable-custom-all-reduce --
ssd-kv-offload-gb 100 --dram-kv-offload 10
```

- VLLM_USE_V1=1：啟用 vLLM V1 Engine
- VLLM_WORKER_MULTIPROC_METHOD=s
pawn：設定 Python Multiprocessing 啟動方式
- python3 vllm_api_server.py：啟動 vLLM API Server
- --model：指定模型路徑，
/app是 Docker Volume 掛載進來的模型資料夾
- --nvme_path：指定 SSD Offload 路徑
- --gpu-memory-utilization：設定 GPU 使用率
- --max-model-len：設定最大 Context Length
- --tensor-parallel-size：多 GPU 設定
- --enable-chunked-prefill：啟用 Chunked Prefill，
將Prompt分段處理
- --enable-prefix-caching：啟用 Prefix Cache，
若多次請求，前半段 prompt 相同，不需要重新計算前面的 KV Cache
- --disable-custom-all-reduce：停用自訂 All Reduce，避免通訊錯誤
- --ssd-kv-offload-gb：設定 SSD KV Cache 空間
- --dram-kv-offload-gb：設定 DRAM KV Cache 空間

Demo aiDAPTIVLink3.0

```
nohup \  
bash -c "VLLM_USE_V1=1 VLLM_WORKER_MULTIPROC_METHOD=spawn\  
python3 vllm_api_server.py\  
--model /app/gemma-3-12b-it\  
--served-model-name gemma-3-12b-it\  
--nvme_path /mnt/nvme0/inference\  
--gpu-memory-utilization 0.8\  
--max-model-len 60000\  
--tensor-parallel-size 2\  
--enable-chunked-prefill\  
--enable-prefix-caching\  
--disable-custom-all-reduce\  
--ssd-kv-offload-gb 100\  
--dram-kv-offload 10" > vllm.log 2>&1 &
```

監看狀態
tail -f vllm.log

查看進程
ps aux | grep vllm_api_server | grep -v grep
殺掉進程
kill <pid>

Demo aiDAPTIVLink3.0

5. 查看Logs

```
root@2d272c6f3064:/home/root/aiDAPTIV2/script# VLLM_USE_V1=1 VLLM_WORKER_MULTIPROC_METHOD=spawn python3 vllm_api_server.py --model /app/Meta-Llama-3.1-8B-Instruct-AWQ-INT4/ --nvme_path /mnt/nvme0/inference --gpu-memory-utilization 0.9 --max-model-len 4096 --tensor-parallel-size 1 --enable-chunked-prefill --enable-prefix-caching --disable-custom-all-reduce --ssd-kv-offload-gb 100 --dram-kv-offload-gb 0
(APIServer pid=75192) INFO 05-25 09:39:02 [api_server.py:1772] vLLM API server version 0.12.0
(APIServer pid=75192) INFO 05-25 09:39:03 [utils.py:253] non-default args: {'model': '/app/Meta-Llama-3.1-8B-Instruct-AWQ-INT4/', 'max_model_len': 4096, 'disable_custom_all_reduce': True, 'enable_prefix_caching': True, 'enable_chunked_prefill': True, 'kv_transfer_config': KVTransferConfig(kv_connector='AidaptivConnectorV1', engine_id='2c4153ee-c213-426e-91b1-e16b7458cc46', kv_buffer_device='cuda', kv_buffer_size=10000000000.0, kv_role='kv_both', kv_rank=None, kv_parallel_size=1, kv_ip='127.0.0.1', kv_port=14579, kv_connector_extra_config={'use_aidaptiv': True, 'offload_path': '/mnt/nvme0/inference', 'ssd_kv_offload_gb': 100, 'dram_kv_offload_gb': 0, 'resume': True, 'lora_module': None}, kv_connector_module_path=None, enable_permute_local_kv=False)}
(APIServer pid=75192) INFO 05-25 09:39:07 [model.py:637] Resolved architecture: LlamaForCausalLM
(APIServer pid=75192) INFO 05-25 09:39:07 [model.py:1750] Using max model len 4096
(APIServer pid=75192) INFO 05-25 09:39:07 [awq_marlin.py:163] The model is convertible to awq_marlin during runtime. Using awq_marlin kernel.
(APIServer pid=75192) INFO 05-25 09:39:07 [scheduler.py:228] Chunked prefill is enabled with max_num_batched_tokens=2048.
(APIServer pid=75192) WARNING 05-25 09:39:07 [vllm.py:896] Turning off hybrid kv cache manager because '--kv-transfer-config' is set. This will reduce the performance of vLLM on LLMs with sliding window attention or Mamba attention. If you are a developer of kv connector, please consider supporting hybrid kv cache manager for your connector by making sure your connector is a subclass of 'SupportsHMA' defined in kv_connector/v1/base.py.
(EngineCore_DP0 pid=75355) INFO 05-25 09:39:11 [core.py:93] Initializing a V1 LLM engine (v0.12.0) with config: model='/app/Meta-Llama-3.1-8B-Instruct-AWQ-INT4/', speculative_config=None, tokenizer='/app/Meta-Llama-3.1-8B-Instruct-AWQ-INT4/', skip_tokenizer_init=False, tokenizer_mode=auto, revision=None, tokenizer_revision=None, trust_remote_code=False, dtype=torch.float16, max_seq_len=4096, download_dir=None, load_format=auto, tensor_parallel_size=1, pipeline_parallel_size=1, data_parallel_size=1, disable_custom_all_reduce=True, quantization=awq_marlin, enforce_eager=False, kv_cache_dtype=
```


Demo aiDAPTIVLink3.0

5. 查看Logs

```
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /finalize, Methods: POST
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /reclaim_storage, Methods: POST
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /ping, Methods: GET
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /ping, Methods: POST
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /invocations, Methods: POST
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /metrics, Methods: GET
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /classify, Methods: POST
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /v1/embeddings, Methods: POST
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /score, Methods: POST
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /v1/score, Methods: POST
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /rerank, Methods: POST
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /v1/rerank, Methods: POST
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /v2/rerank, Methods: POST
(APIServer pid=75192) INFO 05-25 09:41:16 [launcher.py:46] Route: /pooling, Methods: POST
(APIServer pid=75192) INFO: Started server process [75192]
(APIServer pid=75192) INFO: Waiting for application startup.
(APIServer pid=75192) INFO: Application startup complete.
(APIServer pid=75192) [2026-05-25 09:41:16][MDW] [Info] - Starting Phison warm-up.
(EngineCore_DP0 pid=75355) [2026-05-25 09:41:17][MDW] [Info] - Post-initializing aiDAPTIVEngine
(EngineCore_DP0 pid=75355) [2026-05-25 09:41:18][MDW] [Info] - init_kv_cache_buffer: device cpu, num blocks 3730
(EngineCore_DP0 pid=75355) [2026-05-25 09:41:18][MDW] [Info] - DRAM space: 0 GB, KV cache size per block 0.001953125 GB
(EngineCore_DP0 pid=75355) [2026-05-25 09:41:18][MDW] [Info] - # DRAM blocks: 0.0
(EngineCore_DP0 pid=75355) [2026-05-25 09:41:18][MDW] [Info] - SSD space: 100 GB, KV cache size per block 0.001953125 GB
(EngineCore_DP0 pid=75355) [2026-05-25 09:41:18][MDW] [Info] - # SSD blocks: 51200.0
(EngineCore_DP0 pid=75355) [2026-05-25 09:41:42][MDW] [Info] - Storing KV cache for 7 out of 7 tokens (skip_leading_tokens=0) for request cmpl-a030c2a023c8aa0e-0
(APIServer pid=75192) INFO: 127.0.0.1:42474 - "POST /v1/completions HTTP/1.1" 200 OK
(APIServer pid=75192) [2026-05-25 09:41:43][MDW] [Info] - [Warm-up Success]: {'id': 'cmpl-a030c2a02
```


Demo aiDAPTIVLink3.0

6. 測試

```
user@user-Pro-ET500I-W8:~/Desktop$ curl http://127.0.0.1:8000/v1/chat/completions \
-H "Content-Type: application/json" \
-d '{
  "model": "/app/Meta-Llama-3.1-8B-Instruct-AWQ-INT4/",
  "messages": [
    {"role": "user", "content": "hi."}
  ],
  "temperature": 0,
  "max_tokens": 50,
  "top_p": 0.1
}'
{"id": "chatcmpl-9fb9a4cc4f5d29ff", "object": "chat.completion", "created": 1779702519, "model": "/app/Meta-Llama-3.1-8B-Instruct-AWQ-INT4/", "choices": [{"index": 0, "message": {"role": "assistant", "content": "How can I assist you today?"}, "refusal": null, "annotations": null, "audio": null, "function_call": null, "tool_calls": [], "reasoning": null, "reasoning_content": null}, {"logprobs": null, "finish_reason": "stop", "stop_reason": null, "token_ids": null}], "service_tier": null, "system_fingerprint": null, "usage": {"prompt_tokens": 37, "total_tokens": 45, "completion_tokens": 8, "prompt_tokens_details": null}, "prompt_logprobs": null, "prompt_token_ids": null, "kv_transfer_params": null, "metrics": {"finish_reason": 0, "e2e_latency": 0.11920952796936035, "num_prompt_tokens": 37, "num_generation_tokens": 8, "max_tokens_param": 50, "queued_time": 1.0378000297350809e-05, "prefill_time": 0.029447980999975698, "inference_time": 0.11726207799983968, "decode_time": 0.08781409699986398, "mean_time_per_output_token": 0.012544870999980568, "is_corrupted": false}}user@user-Pro-ET500I-W8:~/Desktop$
```

Demo aiDAPTIVLink3.0

6. (測試) Logs 查看

```
(APIServer pid=75192) INFO: 172.17.0.1:60820 - "POST /v1/chat/completions HTTP/1.1" 200 OK
(APIServer pid=75192) INFO 05-25 09:48:46 [loggers.py:236] Engine 000: Avg prompt throughput: 3.7 tokens/
s, Avg generation throughput: 0.8 tokens/s, Running: 0 reqs, Waiting: 0 reqs, GPU KV cache usage: 0.0%, P
refix cache hit rate: 0.0%, External prefix cache hit rate: 72.7%
(APIServer pid=75192) INFO 05-25 09:48:56 [loggers.py:236] Engine 000: Avg prompt throughput: 0.0 tokens/
s, Avg generation throughput: 0.0 tokens/s, Running: 0 reqs, Waiting: 0 reqs, GPU KV cache usage: 0.0%, P
refix cache hit rate: 0.0%, External prefix cache hit rate: 72.7%
```

Demo aiDAPTIVLink3.0 — Lock System

➤ 架構

- multi gpu

```
<offload_path>
├── rank0/
│   ├── doc1/          <-- Lock Folder
│   │   ├── kv_cache.bin  <-- Lock KV Cache
│   │   ├── prefix_tree.bin
│   │   └── model_info.bin
│   ├── doc2/          <-- Unlock Folder
│   │   ├── kv_cache.bin  <-- Unlock KV Cache
│   │   ├── prefix_tree.bin
│   │   └── model_info.bin
│   ├── kv_cache.bin    <-- Dynamic KV Cache
│   ├── prefix_tree.bin
│   └── model_info.bin
├── rank1/
│   ├── doc1/          <-- Lock Folder
│   │   ├── kv_cache.bin  <-- Lock KV Cache
│   │   ├── prefix_tree.bin
│   │   └── model_info.bin
│   ├── doc2/          <-- Unlock Folder
│   │   ├── kv_cache.bin  <-- Unlock KV Cache
│   │   ├── prefix_tree.bin
│   │   └── model_info.bin
│   ├── kv_cache.bin    <-- Dynamic KV Cache
│   ├── prefix_tree.bin
│   └── model_info.bin
```

補充說明：
不同模型會對應不同的 ID 進行管理

```
Doc1/
kv_cache_3bc1ee0a3c366ea3.bin
kv_cache_75b62c7b034f51a2.bin
model_info_3bc1ee0a3c366ea3.bin
model_info_75b62c7b034f51a2.bin
.phison
.phison log/
prefix_tree_3bc1ee0a3c366ea3.bin
prefix_tree_75b62c7b034f51a2.bin
Test1/
```

Demo aiDAPTIVLink3.0 — Lock System

➤ 實作- Create lock folder

Example:

- POST <http://0.0.0.0:8000/v1/chat/completions>

```
curl -s -X POST http://0.0.0.0:8000/v1/chat/completions -H  
"Content-Type: application/json" -d '{  
  "model": "gemma-3-12b-it",  
  "messages": [{"role": "user", "content": "The future of AI is"}],  
  "temperature": 0,  
  "max_tokens": 500,  
  "top_p": 0.1,  
  "offload_folder_name": "Doc1"  
}' | jq .
```

Demo aiDAPTIVLink3.0 — Lock System

➤ 實作 - Get lock folder

Get lock folder

By using this API, you are able to know the “lock” status of each folder.

Example:

- GET http://0.0.0.0:8000/get_lock_folder

```
// server response
[
  {
    "doc1": "lock",
    "doc2": "unlock",
  }
]
```

```
root@2d272c6f3064:/# curl -X GET http://0.0.0.0:8000/get_lock_folder
"{\"lock folder\": [], \"unlock folder\": [\"Test1\", \"Doc1\", \"Doc3\"]}"
```


Demo aiDAPTIVLink3.0 — Lock System

➤ 實作 - Update lock folder

Example:

- POST http://0.0.0.0:8000/update_lock_folder

// Body

```
{  
  "doc1": "unlock",    <-- This will unlock doc1 folder  
}
```

```
root@2d272c6f3064:/# curl -X POST http://0.0.0.0:8000/update_lock_folder -H "Content-Type: application/json" -d '{"Doc1":"lock"}'  
"Folder status updated."root@2d272c6f3064:/#  
root@2d272c6f3064:/# curl -X GET http://0.0.0.0:8000/get_lock_folder  
"{\"lock folder\": [\"Doc1\"], \"unlock folder\": [\"Test1\", \"Doc3\"]}"root@2d272c6f3064:/#
```

Demo aiDAPTIVLink3.0 — Lock System

➤ 實作 - Delete lock folder

Example:

- POST http://0.0.0.0:8000/delete_unlock_folder

```
// Body

{
  "delete_unlock_folder": ["doc1"],  <-- This will delete doc1 folder
}
```

```
root@2d272c6f3064:/# curl -X POST http://0.0.0.0:8000/delete_unlock_folder \
-H "Content-Type: application/json" \
-d '{
  "delete_unlock_folder": ["Doc3"]
}'
"Subfolder deleted."root@2d272c6f3064:/#
```

```
root@2d272c6f3064:/# curl -X GET http://0.0.0.0:8000/get_lock_folder
"{\"lock folder\": [\"Doc1\"], \"unlock folder\": [\"Test1\"]}"root@
```

Demo aiDAPTIVLink3.0 — Lock System

您的使用情境	建議做法	要不要加參數？
一般對話、隨便問問	完全不要加參數	不用，讓系統動態管理就好，省得佔硬碟空間。
確認會有頻繁對話 (Multi-turn)	第一輪加上名稱 (如 doc-1)	Create + offload_folder_name: "Doc1"， 避免對話到一半，前面的 Cache 被其他請求擠掉，導致記憶體錯亂。
想暫時廢棄某個鎖定 或者 準備刪除	呼叫 Update API 改成 unlock	GET /get_lock_folder # 確認 folder lock or not POST /update_lock_folder 搭配 {"Doc1": "unlock"} # or lock 必須先做這步，否則刪不掉
執行刪除	呼叫 Delete API	POST /delete_unlock_folder 搭配 {"delete_unlock_folder": ["Doc1"]}